

Translated

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Strongly connected components and the condensation graph

Definitions

Let $G=(V,E)$ be a directed graph with vertices V and edges $E \subseteq V \times V$. We denote with $n=|V|$ the number of vertices and with $m=|E|$ the number of edges in G . It is easy to extend all definitions in this article to multigraphs, but we will not focus on that.

A subset of vertices $C \subseteq V$ is called a **strongly connected component** if the following conditions hold:

- for all $u,v \in C$, if $u \neq v$ there exists a path from u to v and a path from v to u , and
- C is maximal, in the sense that no vertex can be added without violating the above condition.

We denote with $\text{SCC}(G)$ the set of strongly connected components of G . These strongly connected components do not intersect with each other, and cover all vertices in the graph. Thus, the set $\text{SCC}(G)$ is a partition of V .

Consider this graph G_{example} , in which the strongly connected components are highlighted:

